IKEA GIFTING

Customer Analytics & RFM Segmentation

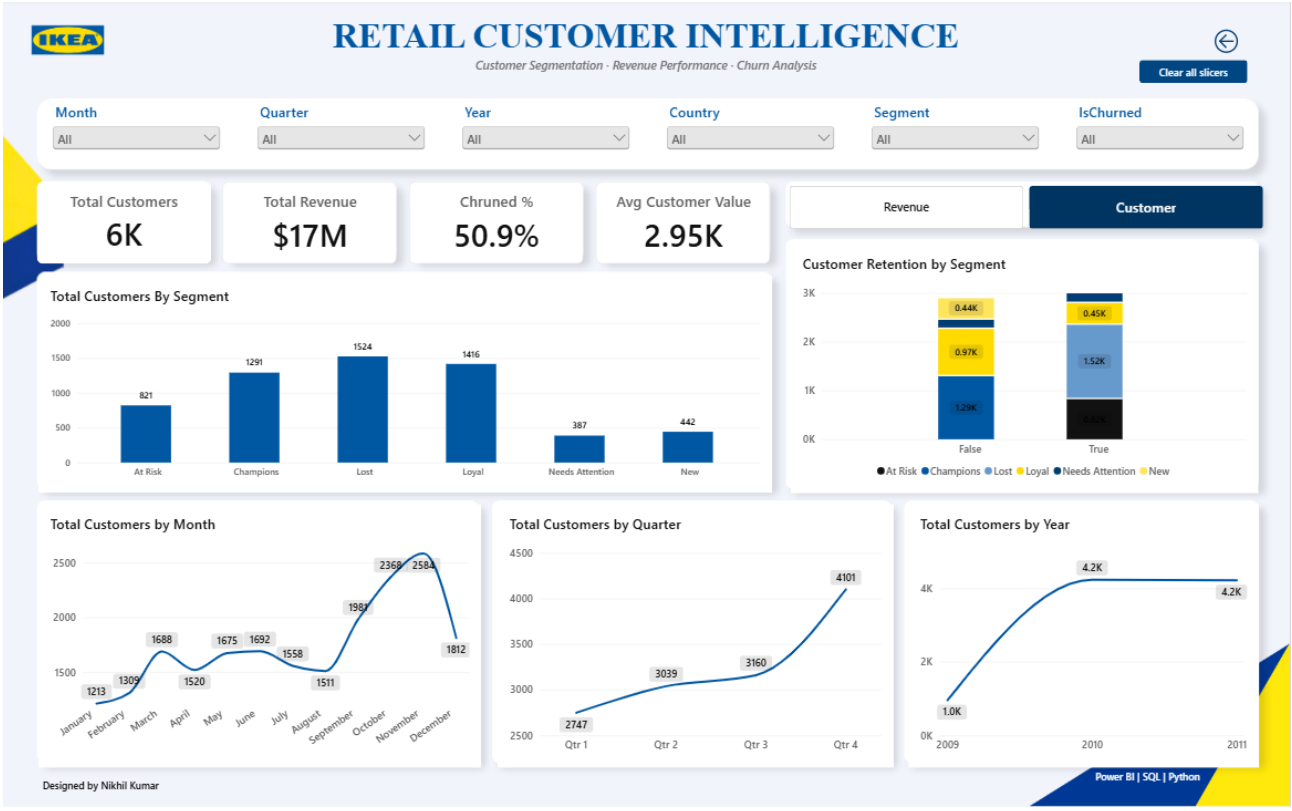
A practical review of customer value, purchasing behaviour, and inactivity-based churn.

Prepared by **Nikhil Kumar** | Portfolio case study | 25 October 2026

1.067M	34,335	5,881	50.9%
RAW TRANSACTIONS	**DUPLICATE ROWS REMOVED**	**CUSTOMERS SCORED**	**INACTIVITY FLAG RATE**
Online Retail II	Exact-row deduplication	RFM customer summary	90-day threshold

The question behind the numbers

Which customers drive the most value, who has gone quiet, and what should the business consider doing next?



Current Overview page - Customer view. The dashboard is preserved as delivered; currency-formatting caveat is documented in this report.

IKEA Gifting is a fictional business name used for this educational portfolio project. This report is not affiliated with the real IKEA corporation.

SECTION 01 / Executive summary

The analysis turns more than one million historical transaction rows into a customer-level view the business can act on.

The business problem was straightforward: leadership could see transactions, but did not have a structured way to compare customer value or identify customers whose purchasing activity had cooled. The project connects a business requirements document, SQL data preparation, Python RFM segmentation, and a Power BI report.

£17.37M	2,991	68.1%	35.5%
POSITIVE-LINE VALUE	**CUSTOMERS FLAGGED**	**CHAMPIONS VALUE SHARE**	**Q4 VALUE SHARE**
Source price units are GBP	50.9% of 5,881	1,291 customers	Months Oct-Dec pooled

What the analysis tells us

- Value is concentrated. Champions represent about 22% of scored customers and account for 68.1% of the eligible transaction value assigned through RFM.
- Inactivity is material. 2,991 customers had not purchased in the 90 days before the 10 December 2011 snapshot date.
- The seasonal pattern is visible. October, November, and December together contribute about 35.5% of the positive-line value when quarters are pooled across the historical period.
- Country averages need context. EIRE ranks first on average country value in the prepared data, but that result comes from just five identified customers.

Recommended management response

Protect the value, then test recovery

Start by protecting Champion relationships and prioritizing a measured reactivation test for At Risk customers. Keep the pilot small, use a holdout group, and judge success on incremental repeat purchases and value rather than campaign clicks alone.

Important interpretation note: the dashboard displays a dollar symbol, but the underlying source prices are GBP-denominated and the workflow does not convert currency. Financial values in this report are therefore described as GBP-denominated source value, with non-positive-quantity rows excluded.

SECTION 02 / Business problem and questions

The BRD is the anchor: each chart should answer a business question, not just fill space.

The business problem

IKEA Gifting was framed as a UK-based, non-store online retailer selling giftware and home decor. The BRD describes a mixed customer base that includes wholesale buyers, which makes customer value harder to interpret. The intended outcome was a more focused view of value and inactivity so Marketing, CRM, Retention, and Business Leadership could prioritize their attention.

Question-to-evidence map

BUSINESS QUESTION	EVIDENCE IN THE PROJECT	STATUS
Who are the most valuable customers?	RFM scores, six segments, segment customer counts and value contribution.	Answered descriptively
Who has gone quiet?	A fixed churn flag: Recency > 90 days at the 10 Dec 2011 snapshot.	Answered by rule
How does performance change over time?	Overview page contains monthly, quarterly, and yearly customer/value views.	Exploratory trends
How does customer value vary by country?	Country performance summary and top-five average customer value view.	Answered, with sample-size caution
What should the business do next?	Segment-specific proposals in this report.	Recommendations proposed; not tested

Scope and boundary

IN SCOPE	NOT IN SCOPE
SQL-based transaction preparation and deduplication	Predictive churn modelling
Python RFM calculation, scoring, and rule-based segments	Real-time or automated customer scoring
A fixed 90-day inactivity flag	Running or measuring live marketing campaigns
Interactive Power BI customer, revenue, segment, and country views	Product-level recommendation engine

Delivery note

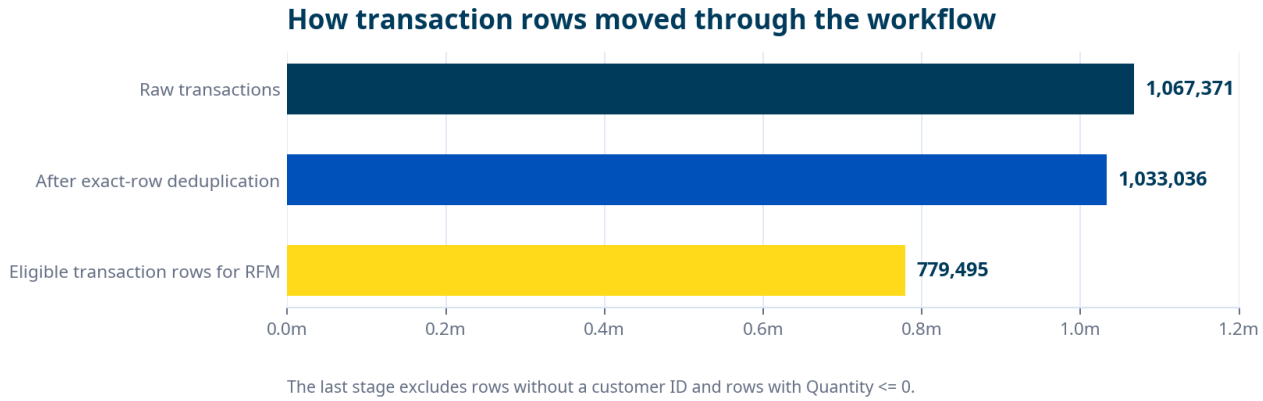
The BRD specified a three-page report including a home/navigation page. The actual PBIX in this repository contains two report pages: Overview and RFM Analysis, with alternative views controlled by bookmarks. This report describes the delivered artifact, not the original page-count plan.

SECTION 03 / Data and quality checks

A useful customer model starts with knowing what made it into the analysis and what did not.

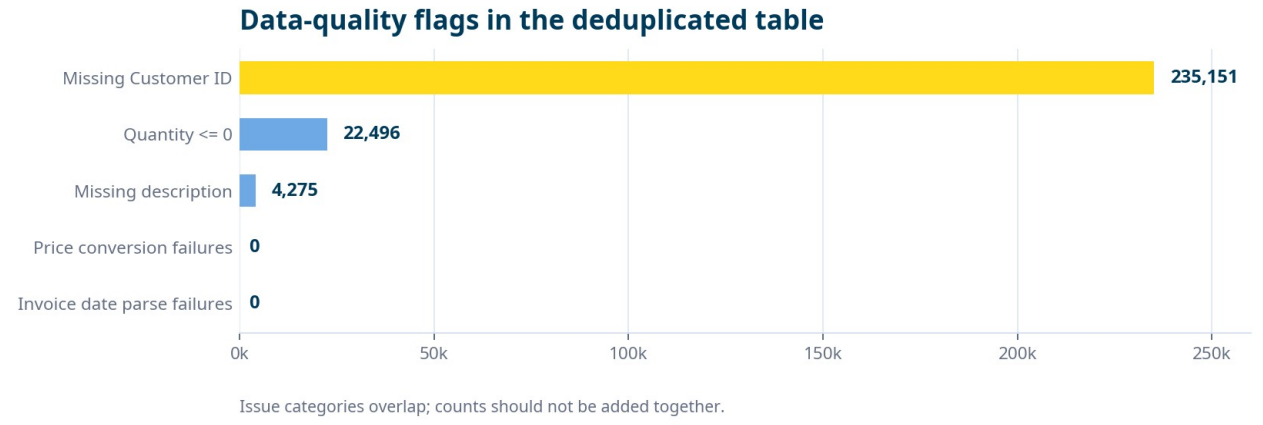
The source is the UCI Online Retail II dataset, containing two annual extracts with invoice, stock code, description, quantity, invoice date, unit price, customer ID, and country. The raw archive contains 1,067,371 rows across the two CSV files.

541,910	525,461	1,033,036	779,495
SOURCE FILE 1	**SOURCE FILE 2**	**AFTER DEDUP**	**RFM INPUT LINES**
CSV rows	CSV rows	SQL output row count	ID present; qty > 0



Record flow is shown only for transaction rows; the 5,881-customer RFM output is a different unit of measure.

Flags observed after deduplication



Data-quality checks were run on the deduplicated table. Issue counts overlap, so they are not additive.

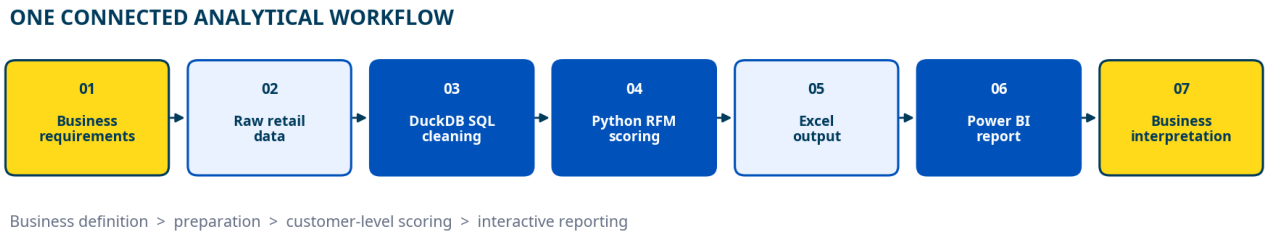
The date parser and price conversion returned no failures in this review of the supplied raw CSV files. That does not make the input universally clean: 235,151 rows in the deduplicated table lacked a customer ID, 22,496 had Quantity <= 0, and 4,275 lacked a description.

What changed for customer analysis

The RFM workflow uses rows with an identified customer and positive quantity. This reduces the deduplicated table from 1,033,036 rows to 779,495 eligible transaction lines. The records with missing IDs and non-positive quantities remain relevant for data-quality and returns analysis, but they are not included in this RFM output.

SECTION 04 / How the analysis was built

The work follows the chain from business definition to repeatable preparation and business-facing reporting.



A single workflow across the BRD, SQL, Python, Excel output, and Power BI.

SQL preparation

- The two annual source tables are combined with UNION ALL and assigned a source-sheet label.
- Quantity, invoice date, price, and customer ID are safely parsed into analytical data types.
- Data-quality checks inspect missing customer IDs, non-positive quantities, missing descriptions, and duplicate transaction attribute combinations.
- ROW_NUMBER() identifies repeated combinations of invoice, item, description, quantity, date, price, customer, and country; one row is retained.

Python customer calculations

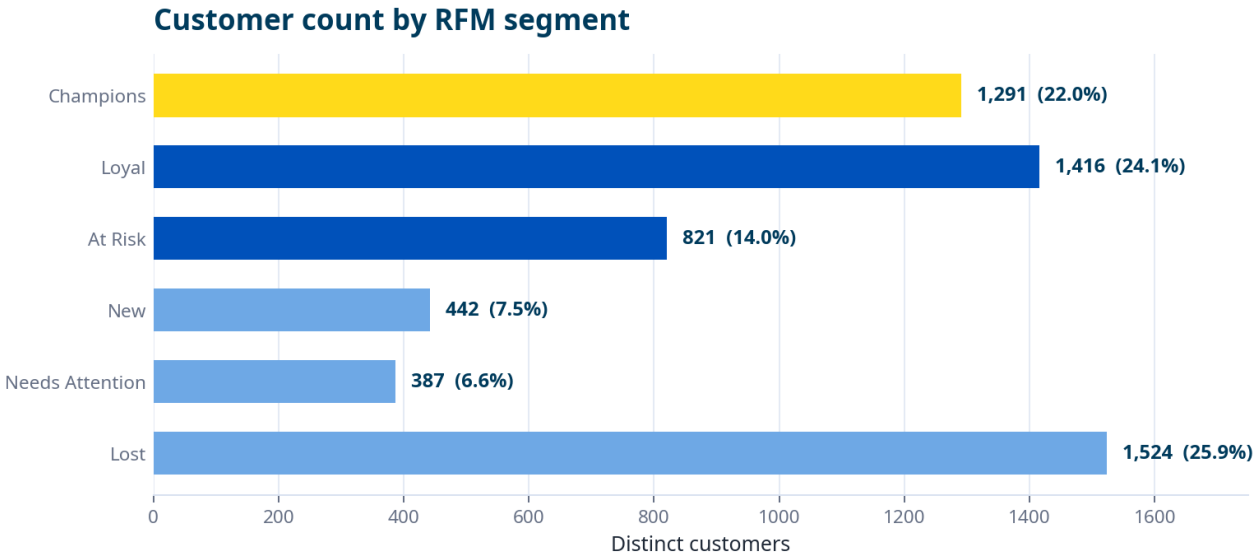
METRIC	CALCULATION USED
Recency	Days between 10 Dec 2011 and the customer's latest purchase date.
Frequency	Count of distinct invoice IDs for the customer.
Monetary	Sum of Quantity × Price for eligible positive-quantity transaction lines.
RFM scoring	Quantile-based score from 1 to 5 for each metric. Recency score is reversed.
Churn flag	IsChurned = Recency > 90 days.

Why the snapshot date matters

The snapshot is fixed at 10 December 2011. If the snapshot date changes, Recency scores, segment assignments, and the inactivity flag can change too. A production workflow would expose this date as a controlled parameter and record it with each refresh.

SECTION 05 / Who are the customers?

RFM groups customers by purchase recency, repeat activity, and value, making the total base easier to interpret.



Customer counts are calculated at the customer ID grain after eligibility filters.

SEGMENT	CUSTOMERS	CUSTOMER SHARE	AVG. VALUE*	FLAGGED
Champions	1,291	22.0%	£9,169	0
Loyal	1,416	24.1%	£1,908	446
At Risk	821	14.0%	£1,932	821
New	442	7.5%	£887	0
Needs Attention	387	6.6%	£529	200
Lost	1,524	25.9%	£429	1,524

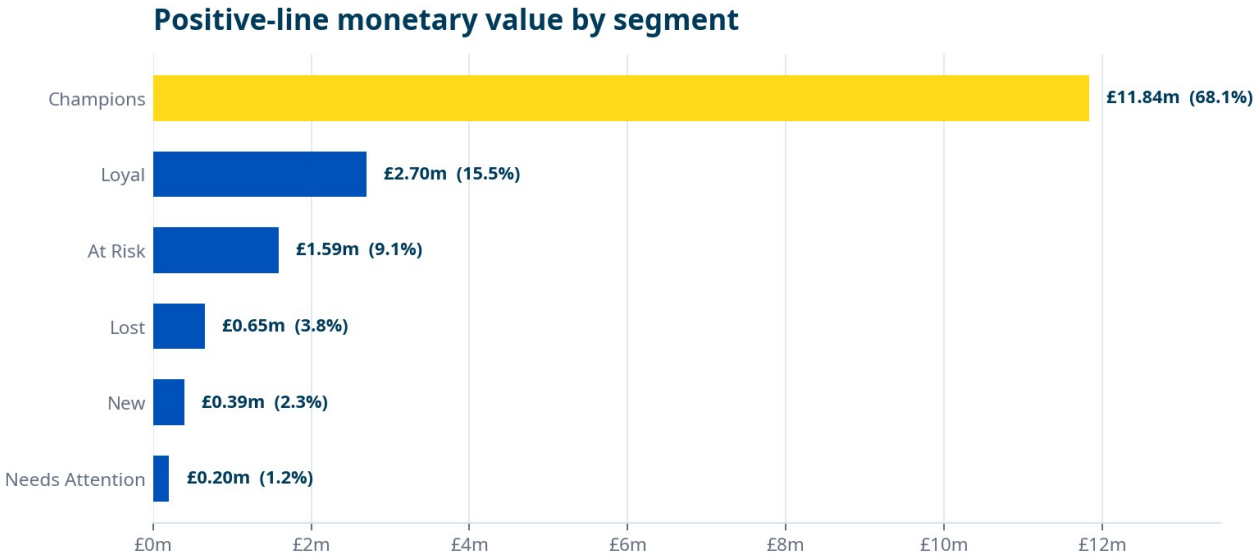
**Average value is the customer-level Monetary total averaged across customers in each segment. Values are GBP-denominated source values; the dashboard currently displays $ symbols.*

Three points worth noting

- Lost is the largest group by count: 1,524 customers, or about 25.9% of the scored base.
- Champions includes 1,291 customers (22.0%). Their individual monetary totals are markedly higher than the other segments.
- At Risk contains 821 customers and has a higher average customer value than Loyal in this output, making it a sensible group for a carefully measured reactivation test.

SECTION 06 / Where does value sit?

The segment view shows a concentrated value profile, while the average alone does not explain every group.



Value is calculated as Quantity × Price after removing non-positive quantities and rows without an identified customer.

£11.84M	£2.70M	£1.59M	£17.37M
CHAMPIONS VALUE	LOYAL VALUE	AT RISK VALUE	TOTAL ANALYSIS VALUE
68.1% of total	15.5% of total	9.1% of total	Eligible positive lines

SEGMENT	VALUE	VALUE SHARE	CUSTOMER SHARE
Champions	£11.84m	68.1%	22.0%
Loyal	£2.70m	15.5%	24.1%
At Risk	£1.59m	9.1%	14.0%
New	£0.39m	2.3%	7.5%
Needs Attention	£0.20m	1.2%	6.6%
Lost	£0.65m	3.8%	25.9%

The mean customer Monetary value is about £2,954, while the median is £866. The mean is more than three times the median, which is consistent with a right-skewed distribution where a relatively small number of high-value customers pull the average upward.

The main finding

Champions represent roughly one in five customers but generate about two-thirds of the eligible transaction value. The implication is not to spend indiscriminately; it is to protect those customer relationships and test retention activity against measurable outcomes.

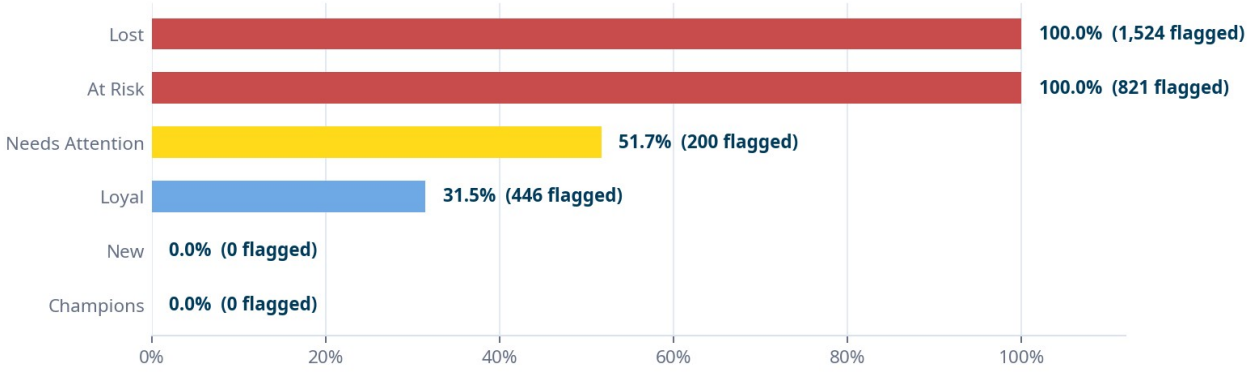
Currency and scope: the total is £17.37m in source-denominated value for identified-customer, positive-quantity lines. It is not net sales after returns, because non-positive quantities were excluded from the RFM input.

SECTION 07 / Who has gone quiet?

The project uses a clear and interpretable inactivity rule, not a predictive model.

2,991	5,881	50.9%	10 DEC 2011
CUSTOMERS FLAGGED	CUSTOMERS SCORED	OVERALL FLAG RATE	SNAPSHOT DATE
Recency > 90 days	All RFM customers	2,991 / 5,881	Historical cut-off

Customers meeting the 90-day inactivity rule



Churn flag = Recency > 90 days at the 10 Dec 2011 snapshot.

Flag rates follow the RFM outputs and the fixed 90-day threshold.

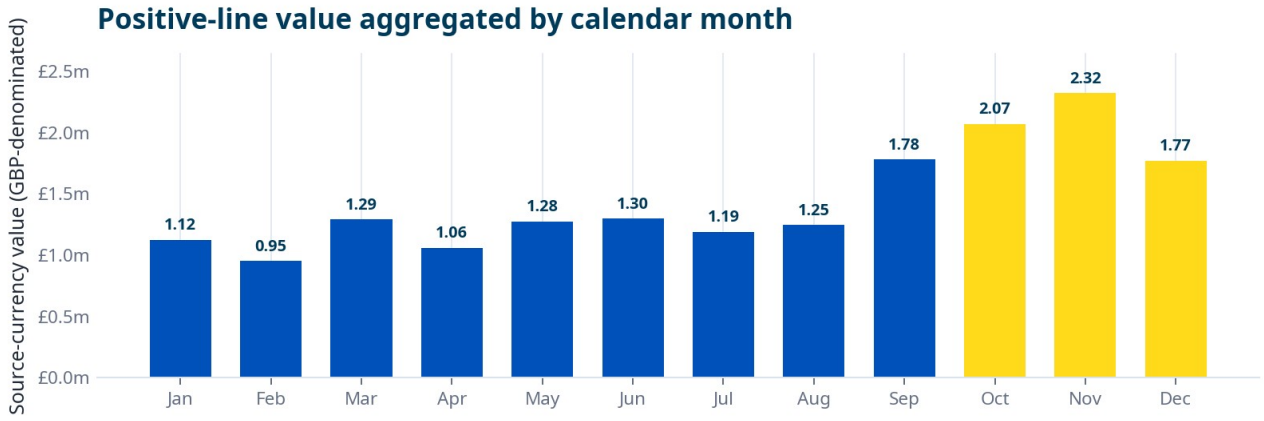
SEGMENT	CUSTOMERS	FLAGGED	FLAG RATE
Champions	1,291	0	0.0%
Loyal	1,416	446	31.5%
At Risk	821	821	100.0%
New	442	0	0.0%
Needs Attention	387	200	51.7%
Lost	1,524	1,524	100.0%

Interpret the flag carefully

A customer with no purchase in the last 90 days is flagged as churned by the project rule. That does not prove the customer is permanently lost or that a campaign can win them back. Before acting, check margin, contactability, prior purchase pattern, and the economics of reactivation.

SECTION 08 / When and where is value recorded?

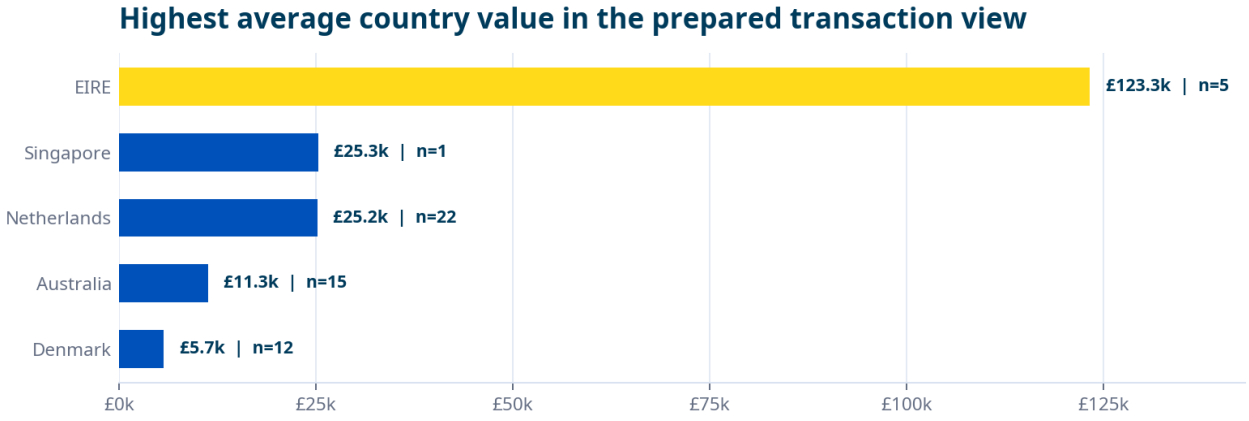
Time and country views add context, but high averages need to be read alongside sample size.



Monthly totals combine the months of the year across the full historical period; they are not a year-over-year comparison.

Month-of-year aggregation pools January values together, February values together, and so on across the historical data.

Seasonal read: October, November, and December together contribute approximately 35.5% of eligible positive-line value across all quarters in the dataset. November is the highest month-of-year total at £2.32m.



Average = country value / distinct identified customers in that country. Small n makes rankings volatile.

Average country value is shown alongside the number of distinct customers identified in that country.

Geographic read: The UK contributes approximately 82.8% of eligible positive-line value. EIRE ranks first on average country value, but the base is only five customers; Singapore has one. High averages from tiny groups should not drive investment decisions without checking customer-level detail.

SECTION 09 / Dashboard evidence / Overview: Customer

This view is designed to answer: how are customers distributed, and what does the inactivity flag look like across segments?

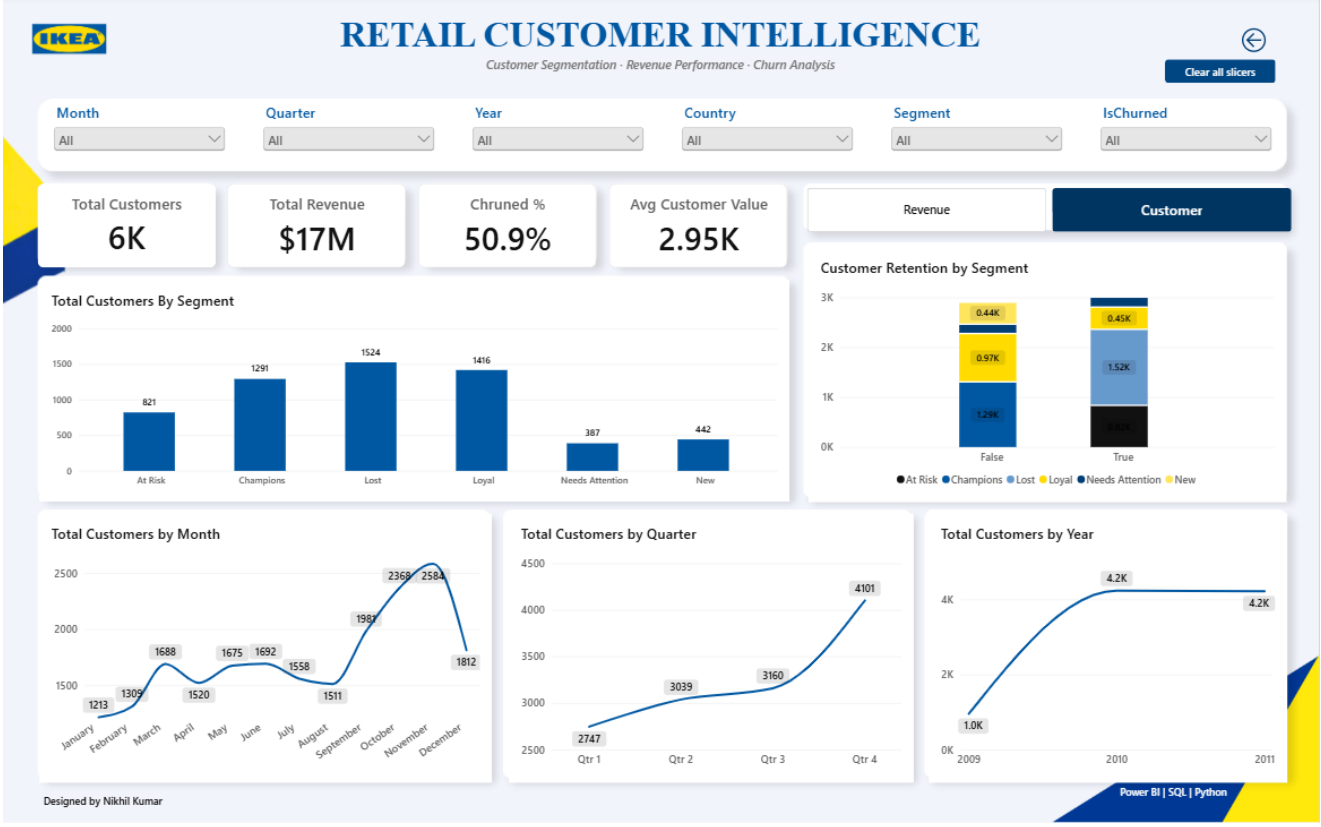


Figure 1. Current Power BI Overview page with Customer selected. The report uses six slicers and includes a customer/revenue bookmark toggle.

What a stakeholder can read here

- The headline cards summarise customers, revenue, churn percentage, and average customer value.
- Customer count by segment shows the size of each RFM group; the retention-by-segment visual compares churn-flag status across segment labels.
- Monthly, quarterly, and yearly visuals provide a quick view of customer activity patterns at different time grains.

Filters available in the report include Month, Quarter, Year, Country, Segment, and IsChurned. They support exploration of the historic data; the screenshots in this report are static and do not carry interactive state.

> **Reporting note**
>
> The dashboard image displays currency with a dollar sign. No USD conversion was found in the SQL or Python workflow. Read the financial cards as currently formatted and use the GBP-denominated calculations documented in this report for analytical interpretation.

SECTION 10 / Dashboard evidence / Overview: Revenue

The revenue bookmark changes the main segment visual and the retention chart while keeping the same report context.



Figure 2. Current Power BI Overview page with Revenue selected.

What this view adds

- The segment bar chart makes revenue contribution visible without requiring the stakeholder to infer value from customer count.
- The retention pie view compares the distribution of value across segments in the current report configuration.
- Monthly, quarterly, and yearly charts let a user explore where value appears across the historical period.

The month labels pool months of the year across the dataset; the November peak is not a single-year November value. Full annual comparisons also need care because 2009 contains only December and 2011 ends on 9 December.

Interpretation guardrail

The dashboard is an exploratory reporting layer. It shows where patterns appear in the history; it does not establish why the pattern happened or whether a marketing action caused it.

SECTION 11 / Dashboard evidence / RFM: Segment

This view brings customer value and segment characteristics together for a closer review.



Segment	Total Customers	Total Revenue	Avg. R_Score	Avg. F_Score	Avg. M_Score
At Risk	821	$15,86,359	1.72	3.92	3.85
Champions	1291	$1,18,36,648	4.71	4.88	4.84
Lost	1524	$6,53,885	1.45	1.56	2.10
Loyal	1416	$27,01,267	3.48	4.00	3.90
Needs Attention	387	$2,04,624	3.00	1.59	2.56
Total	5881	$1,73,74,804	3.99	4.34	4.34

Figure 3. Current Power BI RFM Analysis page with Segment selected.

What this view adds

- The summary table combines customer counts and value with average Recency, Frequency, and Monetary scores.
- Average customer value by segment helps distinguish large customer groups from groups with higher value per customer.
- Churn trends across month, quarter, and year show how the fixed inactivity flag is distributed in the report view.

A modeling note for the next version: RFM scores are created at customer grain and then merged back onto transaction rows. If Power BI averages those repeated scores directly from the transaction table, the result may be transaction-weighted rather than a simple mean across customers. Before operational use, define each score measure at the intended grain and reconcile it to the customer-level RFM table.

SECTION 12 / Dashboard evidence / RFM: Country

Geographic comparisons are useful for exploration, but averages are sensitive to group size.



Figure 4. Current Power BI RFM Analysis page with Country selected.

What this view adds

- Country performance provides a way to compare the value and customer characteristics of historical markets.
- The top-five average value visual helps surface markets with high value per identified customer.
- The table includes total customers, total value, and average RFM scores for the country view.

Averages are not the whole story

In the prepared transaction data, EIRE has approximately £123.3k of value per identified customer across just five customers, while Singapore has about £25.3k across one customer. These are useful prompts for investigation, not proof of a scalable market opportunity.

SECTION 13 / Segment-level action plan

These are proposed actions based on the descriptive analysis. No campaign was executed or tested as part of the project.

SEGMENT	CURRENT SIGNAL	POTENTIAL BUSINESS RESPONSE	MEASURE SUCCESS WITH
Champions	1,291 customers; 68.1% value share; 0% flagged	Protect the relationship. Consider recognition, relevant early access, and service-led retention before broad discounting.	Repeat purchase rate, value retained, margin after incentives
At Risk	821 customers; 9.1% value share; 100% flagged	Prioritize a small, measurable reactivation pilot using historical value and purchase pattern to size the offer.	Reactivation rate, incremental value vs. holdout, cost per recovered customer
Loyal	1,416 customers; 15.5% value share; 31.5% flagged	Encourage the next purchase and monitor the move from active to inactive. Use cadence and relevance rather than blanket offers.	Repeat invoice rate, time to next order, retained value
New	442 customers; 2.3% value share; 0% flagged	Make it easy to reach a second purchase with helpful onboarding or complementary product communication.	Second-purchase conversion, days to second order
Needs Attention	387 customers; 1.2% value share; 51.7% flagged	Use low-cost, relevant reminders and test whether a smaller intervention changes behaviour.	Reactivation, contribution after campaign cost
Lost	1,524 customers; 3.8% value share; 100% flagged	Use a controlled, low-cost win-back test only after checking contactability and expected economics. Do not assume every flagged customer is worth reacquiring.	Incremental recovered value, margin, unsubscribe/complaint rate

A sensible 30-day next step

1. Validate the currency and financial definitions in the dashboard before using the figures outside the portfolio.
2. Choose one segment for a small retention test; document a clear hypothesis, audience, holdout, offer cost, and success metric.
3. Compare incremental behaviour against the holdout, not only clicks or total sales from campaign recipients.
4. Record what worked, what did not, and whether the result justifies a wider rollout.

> **Decision principle**
>
> Use RFM to focus attention. Use a controlled test to learn whether an intervention actually changes behaviour.

SECTION 14 / Limitations and reproducibility

The project is a strong descriptive case study; before a business rollout, the following checks would make the decision trail more robust.

Limitations that affect interpretation

CONSIDERATION	WHY IT MATTERS / NEXT CHECK
Currency display	The source prices are GBP-denominated. The Power BI report displays $ symbols and the SQL/Python workflow does not convert currency. Correct the display or document a real exchange-rate conversion before operational reporting.
Treatment of returns	The RFM input excludes Quantity <= 0. This avoids including returns/cancellation-like lines in customer RFM, but the monetary result is positive-line value, not a net revenue measure after returns.
Fixed churn rule	The 90-day threshold is simple and interpretable, but it is not a trained prediction and may not match different customer purchase cycles.
Snapshot date	All Recency values use 10 Dec 2011. If the snapshot changes, segment and churn assignments may change.
Customer type	Wholesale and individual retail customers are not separated in this analysis.
Average score grain	Scores are calculated at customer grain and then merged back to transactions. Averages in Power BI should be checked to ensure they are calculated at the customer grain if that is what the measure is intended to represent.
Small country samples	Very high averages in countries with only one or a few customers are volatile. Always show sample size with country-level averages.
Historical data	The data covers December 2009 through 9 December 2011. Findings are not a current view of customer behaviour.

How to reproduce the current workflow

1. Download Online Retail II from the UCI Machine Learning Repository and import the two source sheets into DuckDB using the table names expected by the SQL script.
2. Run sql/01_retail_data_cleaning_duckdb.sql to create RetailCombined and RetailCleaned. Because the script uses CREATE TABLE, use a fresh database or remove existing derived tables before rerunning.
3. Export main.RetailCleaned to RetailCleaned.csv and place it in the notebook working directory, or update the notebook path.
4. Run notebooks/ikea_rfm_segmentation.ipynb sequentially. The notebook produces EcommerceDashboardData.xlsx.
5. Open powerbi/IKEA.pbix in Power BI Desktop, update the local source path if required, and refresh.

Data source: Chen, D. (2012), Online Retail II, UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/502/online+retail+ii | DOI: 10.24432/C5CG6D. The dataset is made available under CC BY 4.0; refer to the UCI page for license and attribution details.

Business disclaimer: IKEA Gifting is a fictional name used for an educational portfolio project and is not affiliated with or representative of the real IKEA corporation.

Portfolio assessment: what this work demonstrates

Strong foundation; clear next upgrades

The project demonstrates an end-to-end descriptive analytics workflow: business questions, SQL preparation, Python customer scoring, and Power BI communication. Before using the figures for a real management decision, the next priorities are to correct the currency display, explicitly define returns treatment, validate score measures at customer grain, and make the SQL script safe to rerun.